

id – идентификатор кредитной заявки

rn – номер кредитной заявки в истории кредитов, num

pre_since_opened –

Feature: pre_since_opened

Group: credit_history / temporal

Origin: numerical (days since credit opened)

Encoding: binned into 20 ordinal categories

AUC: **0.50**

Default rate: no monotonic trend, noisy

Decision: drop (weak signal)

Comment: Heavy binning destroyed original temporal signal.

pre_since_confirmed –

Feature: pre_since_confirmed

Group: credit_history / temporal

Origin: numerical (days since information confirmation)

Encoding: binned into 17 ordinal categories

AUC: **0.51**

Default rate: weak, non-monotonic, low variance

Decision: drop (very weak signal)

Comment:

Administrative temporal feature with minimal predictive power.

Slightly above noise, but not worth keeping in current binned form.

pre_pterm –

Feature: pre_pterm

Group: credit_history / temporal

Origin: numerical (planned days from credit opening to closure)

Encoding: binned into 18 ordinal categories

AUC: **0.5225** (top-15 among 62)

Default rate: non-monotonic but several noticeable spikes

Decision: keep / monitor

Comment:

Temporal feature reflects the planned credit term; binning introduces some noise, but the signal is relatively weak, though stronger than previous "pre_since_*" features.

Important for the model: clients with rare long terms (e.g., bin 5 → 16.7% default) produce local risk spikes.

pre_fterm –

Feature: pre_fterm

Group: credit_history / temporal

Origin: numerical (actual days from credit opening to closure)

Encoding: binned into 17 ordinal categories

AUC: **0.5161** (top-28 among 62)

Default rate: non-monotonic with minor local spikes

Decision: keep / monitor

Comment:

Temporal feature reflects the actual credit term; binning introduces some noise, signal is weak but slightly above random. Rare or extreme bins (e.g., bin 3 → 6.1% default) create local risk spikes.

БИН 4 – 1 клиент.

pre_till_pclose –

Feature: pre_till_pclose

Group: credit_history / temporal

Origin: numerical (planned days from data collection to credit closure)

Encoding: binned into 17 ordinal categories

AUC: **0.5005** (top-54 among 62)

Default rate: essentially flat, no clear trend

Decision: drop (very weak signal)

Comment:

Temporal feature representing the planned remaining term until credit closure; binning destroys almost all predictive power.

Default rates fluctuate minimally across bins, producing no usable signal for the model.

Not recommended for inclusion in its current form.

pre_till_fclose –

Feature: pre_till_fclose

Group: credit_history / temporal

Origin: numerical (actual days from data collection to credit closure)

Encoding: binned into 16 ordinal categories

AUC: **0.5061** (top-43 among 62)

Default rate: non-monotonic with minor local fluctuations

Decision: keep / monitor

Comment:

Temporal feature representing the actual remaining term until credit closure; signal is very weak but slightly above random. Some bins (e.g., bin 2 → 4.9% default) show minor spikes, but overall contribution to the model is minimal. Consider merging rare bins for stability.

pre_loans_credit_limit –

Feature: pre_loans_credit_limit

Group: credit / financial

Origin: numerical (credit limit)

Encoding: binned into 20 ordinal categories

AUC: **0.5203** (top-24 among 62)

Default rate: non-monotonic with several local spikes

Decision: keep / monitor

Comment:

Feature represents the credit limit of the client; binning introduces some noise but preserves minor predictive signal.

Certain bins (e.g., bin 18 → 5.2% default, bin 13 → 4.3%) indicate elevated risk.

Useful for the model when combined with other financial features, though no clear monotonic relationship exists.

pre_loans_next_pay_summ –

Feature: pre_loans_next_pay_summ

Group: credit / payment

Origin: numerical (next scheduled payment amount)

Encoding: binned into 7 ordinal categories

AUC: **0.5004** (top-55 among 62)

Default rate: flat with weak local fluctuations

Decision: drop / very low priority

Comment:

Feature reflects the amount of the next scheduled loan payment.

Default rates across bins are almost uniform, with only minor local variations (e.g., bins 3 and 6).

The feature provides nearly no standalone predictive signal and behaves close to random noise.

pre_loans_outstanding –

Feature: pre_loans_outstanding

Group: credit / balance

Origin: numerical (remaining outstanding loan amount)

Encoding: binned into 5 ordinal categories

AUC: **0.5112** (top-34 among 62)

Default rate: weakly increasing with higher bins

Decision: monitor / low priority

Comment:

The feature captures the remaining unpaid loan balance at the observation date.

Default rate shows a mild upward trend toward higher outstanding amounts, with the highest bin

exhibiting elevated risk. Overall signal is weak but not purely random.

pre_loans_total_overdue –

Feature: pre_loans_total_overdue

Group: credit / delinquency

Origin: numerical (current overdue amount)

Encoding: constant (single bin only)

AUC: **0.5000**

Default rate: undefined (no variation)

Decision: drop

Comment:

The feature has zero variance: all observations fall into a single category.

It carries no information and cannot contribute to model performance.

pre_loans_max_overdue_sum –

Feature: pre_loans_max_overdue_sum

Group: credit / delinquency

Origin: numerical (maximum overdue amount)

Encoding: binned into 3 ordinal categories

AUC: **0.5051** (top-44 among 62)

Default rate: weak non-monotonic pattern

Decision: keep (low priority) / monitor

Comment:

Captures historical delinquency severity, but heavy binning compresses the signal.

Default rate is higher for non-zero bins, yet overall discriminatory power is close to random.

pre_loans_credit_cost_rate –

Feature: pre_loans_credit_cost_rate

Group: loan_terms / pricing

Origin: numerical (total cost of credit / effective rate)

Encoding: binned into 14 ordinal categories

AUC: **0.5046** (top-45 among 62)

Default rate: weak, non-monotonic with isolated spikes

Decision: keep (low priority) / monitor

Comment:

Represents overall loan pricing; heavy binning flattens the risk signal.

A few rare bins show elevated default rates, but overall discriminatory power is close to random.

pre_loans5 –

Feature: pre_loans5

Group: credit / delinquency

Origin: numerical (number of past delinquencies up to 5 days)

Encoding: heavily binned, sparse ordinal categories

AUC: **0.5019** (top-50 among 62)

Default rate: extremely sparse, unstable spikes

Decision: keep as part of delinquency block / weak individually

Comment:

Captures short-term historical payment delays. Signal is highly compressed:

most observations fall into a single bin, making individual predictive power very weak.

pre_loans530 –

Feature: pre_loans530

Group: credit / delinquency

Origin: numerical (number of past delinquencies from 5 to 30 days)

Encoding: heavily binned, sparse ordinal categories

AUC: **0.5037** (top-47 among 62)

Default rate: highly unstable due to extreme sparsity

Decision: keep as part of delinquency block / very weak individually

Comment:

Captures short-to-mid term payment delays. Most observations collapse into a single bin,

making the standalone signal negligible. Extreme default rates appear only in tiny bins.

pre_loans3060 –

Feature: pre_loans3060

Group: credit / delinquency

Origin: numerical (number of past delinquencies from 30 to 60 days)

Encoding: extremely sparse binned ordinal categories

AUC: **0.4996** (top-58 among 62)

Default rate: meaningless at global level due to near-zero variance

Decision: keep as part of delinquency block / drop if isolated

Comment:

Represents mid-term historical delinquencies. Almost all observations collapse into a single bin,

making the feature effectively non-informative on its own.

pre_loans6090 –

Feature: pre_loans6090

Group: credit / delinquency

Origin: numerical (number of past delinquencies from 60 to 90 days)

Encoding: binary / near-constant binned feature

AUC: **0.499974** (top-57 among 62)

Default rate: no meaningful variation

Decision: keep as part of delinquency block / negligible standalone value

Comment:

Represents severe historical delinquencies. Almost the entire sample falls into a single bin,

rendering the feature effectively non-informative on its own.

pre_loans90 –

Feature: pre_loans90

Group: credit / delinquency

Origin: numerical (number of past delinquencies over 90 days)

Encoding: pseudo-binary / near-constant categorical

AUC: **0.4995** (top-59 among 62)

Default rate: unstable, driven by extremely rare values

Decision: exclude or recode; unusable in current form

Comment:

Intended to represent very severe (90+ days) historical delinquencies.

However, the feature is almost constant (value 8 for ~99.94% of observations), with only a handful of rare alternative codes.

is_zero_loans5 –

Feature: is_zero_loans_5

Group: credit / delinquency

Origin: binary flag (no past delinquencies up to 5 days)

Encoding: boolean / binary indicator

AUC: **0.5109** (top-35 among 62)

Default rate: lower default rate when flag = 1

Decision: keep as meaningful behavioral stability indicator

Comment:

Indicates complete absence of short-term delinquencies in credit history.

Unlike raw delinquency counts, the zero-flag cleanly separates stable borrowers from those with at least one minor payment delay, providing usable signal.

is_zero_loans530 –

Feature: is_zero_loans_530

Group: credit / delinquency

Origin: binary flag (no past delinquencies from 5 to 30 days)

Encoding: boolean / binary indicator

AUC: **0.5204** (top-23 among 62)

Default rate: meaningfully lower when flag = 1

Decision: keep as strong behavioral credit feature

Comment:

Captures absence of medium-term payment delays, which are more informative than

short technical delays. Provides one of the strongest standalone signals among delinquency zero-flags and clearly separates disciplined borrowers from risky ones.

is_zero_loans3060 –

Feature: is_zero_loans_3060

Group: credit / delinquency

Origin: binary flag (no past delinquencies from 30 to 60 days)

Encoding: boolean / binary indicator

AUC: **0.5146** (top-29 among 62)

Default rate: significantly higher when flag = 0

Decision: keep as important risk marker within delinquency block

Comment:

Indicates absence of serious mid-term delinquencies (30–60 days), which are strong

behavioral signals of financial distress. While relatively rare, their presence sharply increases default risk, making the zero-flag informative despite imbalance.

is_zero_loans6090 –

Feature: is_zero_loans_6090

Group: credit / delinquency

Origin: binary flag (no past delinquencies from 60 to 90 days)

Encoding: boolean / binary indicator

AUC: **0.5094** (top-39 among 62)

Default rate: markedly higher when flag = 0

Decision: keep as supporting risk indicator within delinquency block

Comment:

Captures the absence of very severe historical delinquencies (60–90 days).

Although such events are rare, their presence corresponds to a sharp increase in default probability, making the feature informative in risk segmentation despite moderate standalone AUC.

is_zero_loans90 –

Feature: is_zero_loans90

Group: credit / delinquency

Origin: binary flag (no past delinquencies over 90 days)

Encoding: boolean / binary indicator

AUC: **0.5107** (top-36 among 62)

Default rate: extremely higher when flag = 0

Decision: keep as critical high-severity risk indicator

Comment:

Indicates absence of the most severe historical delinquencies (>90 days).

Such events are rare but correspond to a dramatic increase in default probability, making the feature essential for identifying extreme-risk borrowers despite moderate standalone AUC.

pre_util –

Feature: pre_util

Group: credit / utilization

Origin: numerical (outstanding loan amount relative to credit limit)

Encoding: binned into 20 ordinal categories

AUC: **0.5122** (top-33 among 62)

Default rate: non-monotonic, local spikes at high utilization bins

Decision: keep / monitor

Comment:

Reflects how much of a borrower's credit line is currently used. Higher bins correspond

to clients with a large portion of their credit drawn, which occasionally leads to noticeable increases in default probability. The feature captures utilization risk

and complements other credit exposure indicators like `pre_loans_outstanding` and `pre_loans_credit_limit`.

`pre_over2limit` –

Feature: `pre_over2limit`

Group: credit / delinquency ratio

Origin: numerical (current overdue amount relative to credit limit)

Encoding: binned into 20 ordinal categories

AUC: **0.4991** (bottom-3 among 62)

Default rate: highly unstable, few clients in high bins cause extreme spikes

Decision: drop or use only as auxiliary / experimental

Comment:

Captures the share of credit currently overdue relative to the credit limit. The feature is extremely sparse in high-risk bins, leading to erratic default rates. Most of the population falls into low or zero-overdue bins, providing almost no signal. Extreme spikes are based on very few observations, making this feature unreliable on its own.

`pre_maxover2limit` –

Feature: `pre_maxover2limit`

Group: credit / delinquency ratio

Origin: numerical (maximum historical overdue relative to credit limit)

Encoding: binned into 20 ordinal categories

AUC: **0.5065** (top-42 among 62)

Default rate: spikes in high bins; most clients have low or zero ratios

Decision: keep as auxiliary within delinquency / credit exposure block

Comment:

Represents the ratio of the largest observed overdue amount to the credit limit. While mostly low for the majority, the rare high-overdue cases generate clear local increases in default risk, giving some signal despite overall low prevalence.

`is_zero_util` –

Feature: `is_zero_util`

Group: credit / utilization

Origin: binary flag (remaining unpaid amount to credit limit ratio equals 0)

Encoding: boolean indicator (0 / 1)

AUC: **0.5142** (top-30 among 62)

Default rate: slightly lower when flag = 1

Decision: keep as supporting utilization stability indicator

Comment:

Identifies clients with zero credit utilization relative to their limit.

Although the difference in default rates is small, the flag consistently points to a more stable credit behavior pattern and adds weak but clean signal to the utilization feature block.

is_zero_over2limit – **ФИЧА УЧИТЫВАЕТ ЗАКРЫТЫЕ ПРОСРОЧКИ???**

Feature: is_zero_over2limit

Group: credit / delinquency / utilization

Origin: binary flag (current overdue amount to credit limit ratio equals 0)

Encoding: boolean indicator (0 / 1)

AUC: **0.5017** (top-51 among 62)

Default rate: marginally lower when flag = 1

Decision: keep only as auxiliary feature inside delinquency block

Comment:

Indicates absence of any current overdue debt relative to the credit limit.

The standalone predictive power is extremely weak, but the feature provides structural information about the current delinquency state and may interact with more expressive delinquency variables.

is_zero_maxover2limit –

Feature: is_zero_maxover2limit

Group: credit / delinquency / utilization

Origin: binary flag (maximum overdue amount to credit limit ratio equals 0)

Encoding: boolean indicator (0 / 1)

AUC: **0.5124** (top-31 among 62)

Default rate: noticeably higher when flag = 0

Decision: keep as supporting risk feature within delinquency history block

Comment:

Captures whether the borrower has ever had any overdue amount relative to the credit limit. While the feature is binary and coarse, it reflects the presence of at least one historical delinquency episode, which materially increases default risk.

Особенно полезна:

- в деревьях (разделение «чистых» и «нечистых» историй),
- в комбинации с:
 - `pre_maxover2limit`,
 - `enc_paym_*`.

`enc_loans_account_holder_type` – **МБ СДЕЛАТЬ ФЛАГ НА ЧАСТОЕ ЗНАЧЕНИЕ И ОСТАЛЬНЫЕ**

Feature: `enc_loans_account_holder_type`

Group: credit / account structure

Origin: categorical (type of relationship to the loan)

Encoding: label / ordinal encoding (encoded categories)

AUC: **0.5026** (top-49 among 62)

Default rate: varies across rare categories, unstable

Decision: keep cautiously; low standalone value, potential interaction feature

Comment:

Encodes the borrower's role or relationship to the loan (e.g., primary borrower, co-borrower, guarantor). The dominant category overwhelms the distribution, while smaller categories show higher default rates but suffer from extreme sparsity, making the standalone signal weak and noisy.

`enc_loans_credit_status` –

Feature: `enc_loans_credit_status`

Group: credit / loan lifecycle

Origin: categorical (current credit status)

Encoding: label / ordinal encoding (encoded statuses)

AUC: **0.5102** (top-37 among 62)

Default rate: increases for adverse / rare statuses, unstable due to sparsity

Decision: keep as contextual risk indicator; useful in interactions

Comment:

Represents the current lifecycle status of the credit (e.g., active, closed, delinquent,

written-off). While most loans fall into a small number of “normal” statuses, rare adverse statuses show sharply higher default rates but suffer from low support,

limiting standalone predictive power.

enc_loans_credit_type –

Feature: enc_loans_credit_type

Group: credit / product characteristics

Origin: categorical (type of credit product)

Encoding: label encoding (encoded credit categories)

AUC: **0.5300** (top-11 among 62; top-2 excluding enc_paym_* block)

Default rate: materially varies across credit types

Decision: keep as a core structural risk feature

Comment:

Encodes the type of credit product (e.g., consumer loan, credit card, installment loan, etc.).

Different credit products imply fundamentally different risk profiles and borrower behavior.

The feature shows stable separation between low-risk and high-risk product categories,

making it one of the strongest non-temporal predictors in the dataset.

enc_loans_account_cur –

Feature: enc_loans_account_cur

Group: credit / contract characteristics

Origin: categorical (currency of the credit contract)

Encoding: label encoding (encoded currency categories)

AUC: **0.5010** (top-52 among 62)

Default rate: minimal variation, unstable on rare categories

Decision: keep for completeness; negligible standalone value

Comment:

Encodes the currency in which the credit is denominated.

The feature is heavily dominated by a single currency,
with other currencies appearing extremely rarely.

As a result, it carries almost no discriminative power on its own.

pclose_flag –

Feature: pclose_flag

Group: credit / contract lifecycle

Origin: binary flag (planned close date is undefined)

Encoding: boolean / binary indicator

AUC: **0.5037** (top-48 among 62)

Default rate: slightly higher when flag = 1

Decision: keep as weak auxiliary contract-structure signal

Comment:

Indicates that the planned duration of the credit (from opening to closing)
is not specified in the contract data.

Such cases may correspond to revolving products or contracts
with flexible or open-ended terms, but the standalone predictive
power of the flag is very limited.

fclose_flag –

Feature: fclose_flag

Group: credit / contract lifecycle

Origin: binary flag (actual close date is undefined)

Encoding: boolean / binary indicator

AUC: **0.5080** (top-40 among 62)

Default rate: higher when flag = 1

Decision: keep as auxiliary lifecycle risk indicator

Comment:

Indicates that the actual duration of the credit (from opening to closing) cannot be determined at the observation date.

This typically corresponds to open or still-active loans and reflects incomplete lifecycle information rather than borrower behavior.

rn –

Feature: rn

Group: credit / product sequencing

Origin: numerical (sequence number of credit products in history)

Encoding: ordinal / sequential

AUC: **0.5042** (top-46 among 62)

Default rate: slowly increasing with higher rn

Decision: keep as minor sequential indicator / contextual feature

Comment:

Represents the chronological order of credit products in a client's history.

Higher values correspond to more recent credits.

Alone, the predictive power is very low, but it may provide context when combined with delinquency patterns or credit type changes.

encpaym 0 – 23

enc_paym_2 –

Feature: enc_paym_2

Group: payment history / monthly status

Origin: categorical (coded monthly payment status)

Encoding: integer codes (0,1,2,3)

AUC: **0.550506** (top-1 among 62 features)

Default rate / flag rate:

0 — 0.0295

1 — 0.0464

2 — 0.0516

3 — 0.0498

Decision: keep as key monthly payment indicator

Comment:

Represents the status of the payment in the second month after credit issuance.

The code is categorical, encoding different levels of payment behavior
(e.g., on-time, delayed, partial, missed — exact mapping unknown).

High AUC indicates strong predictive power for default risk.

enc_paym_3 –

Feature: enc_paym_3

Group: payment history / monthly status

Origin: categorical (coded monthly payment status)

Encoding: integer codes (0,1,2,3)

AUC: **0.548414** (top-2 among 62 features)

Default rate / flag rate:

0 — 0.0294

1 — 0.0452

2 — 0.0491

3 — 0.0463

Decision: keep as key monthly payment indicator

Comment:

Represents the status of the payment in the third month after credit issuance.

The code is categorical, encoding different levels of payment behavior (e.g., on-time, delayed, partial, missed — exact mapping unknown).

High AUC indicates strong predictive power for default risk.

enc_paym_* —

Feature: enc_paym_{0..24}

Group: credit / payment / monthly payment status

Origin: categorical (monthly payment status over the last 24 months, encoded)

Encoding: ordinal categorical with 4 levels (0–3), where higher values indicate worse payment status

AUC: 0.523–0.550 for individual features, top-1–top-10 among 62 features; aggregated features show AUC up to 0.560

Default rate: varies across values, with higher status values generally corresponding to higher default probability

Decision: keep selective individual months or use aggregated indicators; optional aggregation recommended for simpler models

Comment:

The enc_paym_* features represent the status of monthly payments for a credit over the last N months (0 = on-time or good, 3 = missed/late or worst).

Individually, early months (e.g., enc_paym_2, enc_paym_3, enc_paym_4) provide the strongest predictive signal, with top AUC values, while later months show progressively lower individual impact.

Default rate analysis indicates that even moderate delinquency (status 1–2) can increase risk, and the jump from 0 to 1 is often the most informative.

Strong correlations exist between adjacent months, making individual features partially redundant in ensemble models like CatBoost.

Aggregated features — e.g., enc_paym_bad_count (number of months with status >0), enc_paym_weighted_score (weighted sum of statuses across months) — capture the overall payment behavior and achieve higher predictive power in simpler models such as logistic regression, while slightly duplicating signal in tree-based models.

Maximum status (enc_paym_max_status) and any bad month flag (enc_paym_any_bad) can be used as coarse indicators, but are generally weaker than frequency- or weight-based aggregates.

These features are particularly useful for modeling the dynamics of credit repayment behavior over time, capturing both the frequency and severity of missed payments.

Decision strategy:

- For complex, nonlinear models (CatBoost, XGBoost): include selective top months or full set, optional aggregated features for interpretability.

- For linear or simple models: prefer aggregated features (bad_count, weighted_score) to reduce dimensionality and enhance signal.